Uday Radadiya

AI/ML Engineer

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PROFESSIONAL SUMMARY

Applied Research Engineer and AI/ML Engineer specializing in agent environments, synthetic task data, and evaluation systems, with experience across generative AI, NLP, and computer vision. Builds reproducible benchmarks, automated verifiers, and hybrid grading pipelines to assess agent reasoning, tool use, and task completion. Combines Python engineering with enterprise experience in RAG, conversational AI, and model evaluation. Skilled in diagnosing agent failures, strengthening data quality, and developing reliable feedback for agent training and evaluation.

WORK EXPERIENCE

Applied Research Engineer, *Abundant (YC-backed)* 01/2026 – 09/2026

- Built evaluation harnesses and benchmark systems for frontier AI coding agents, assessing reasoning, tool use, multi-step debugging, and real-world code execution.
- Authored Harbor-format tasks across data science, ML engineering, causal inference, and distributed systems, with Docker environments, reference solutions, and automated tests.
- Developed synthetic task datasets with controlled variation and realistic data-quality challenges, expanding scenario coverage and supporting evaluation on held-out inputs.
- Engineered outcome-based verifiers and reward checks to assess correctness, accept alternative valid solutions, and reject incorrect outputs and grading shortcuts.
- Integrated deterministic validation with LLM-based grading to evaluate numerical correctness, generated artifacts, and qualitative output quality.
- Built repeatable validation workflows using reference solutions, negative controls, hidden variants, and controlled reruns to expose brittle solutions and grading defects.
- Analyzed agent trajectories and multi-agent results to refine difficulty, detect leakage risks, and distinguish agent mistakes from environment or verifier failures.

Sr. Al/ML Engineer, *DRC SYSTEMS* 03/2025 – 12/2025

- Delivered 5+ end-to-end Al Proofs-of-Concept (POCs) for enterprise clients, showcasing the business value of Al in diverse domains.
- Built and optimized Retrieval-Augmented Generation (RAG) pipelines tailored for diverse enterprise use cases, enhancing response relevance and latency.
- Developed advanced image preprocessing workflows using edge detection and the Segment Anything Model (SAM) to significantly boost model accuracy to 90%+ in vision-based applications.
- Iteratively optimized prompts and fine-tuned LLM interactions to improve response quality, consistency, and task-specific performance.

Key Projects & Responsibilities:

- **Enterprise RAG & Knowledge Retrieval System —** Owned the full ML stack: embedding model selection and benchmarking, chunking strategy experiments, vector index design, hybrid retrieval, re-ranking, and prompt orchestration. Built a custom evaluation harness measuring retrieval precision, answer faithfulness, and latency to guide every design decision quantitatively rather than anecdotally.

- **Computer Vision Intelligence Pipeline —** Led model development end-to-end — exploratory data analysis on raw imagery, preprocessing pipeline design, SAM-based segmentation integration, error analysis, and metric-driven iteration. Ran ablation studies isolating each preprocessing stage's contribution to final accuracy and shipped the configuration that exceeded the 90% production threshold.
- **LLM-Powered Automation Layer —** Owned the modeling and evaluation layer for a multi-step reasoning product — prompt architecture, function-calling schemas, output validators, and offline eval datasets. Built golden test sets and regression suites that caught model behavior drift before deployment.

Al Engineer, *Intralign.ai* 01/2024 – 02/2025

- Integrated OpenAI APIs and optimized prompts to enhance output quality by reducing errors and inconsistencies by 27%.
- Designed, developed, and deployed Rasa-based chatbots for 20+ business use cases, implementing custom responses to improve user engagement, satisfaction and Latency issues.
- Assisted the ML team in creating high-quality datasets and training machine learning models, contributing to a 20% improvement in models accuracy.
- Implemented + Al agents using frameworks like FlowiseAl and LangChain, streamlining workflows and enabling complex, multi- step reasoning.

Key Projects & Responsibilities (Intralign)

- **Conversational AI Platform —** Owned the Rasa modeling stack across 20+ verticals — intent and entity schema design, NLU pipeline configuration, dialogue policy tuning, and custom action development. Built instrumentation to surface low-confidence predictions and misclassifications, feeding them back into labeled retraining sets to drive measurable NLU accuracy gains.
- **Prompt Engineering & LLM Integration Framework —** Built a reusable prompt-evaluation framework on top of OpenAI APIs — golden datasets, automated scoring, regression tests for prompt changes — that made model behavior measurable and reproducible rather than anecdotal. Authored internal prompt-pattern guidelines adopted across the engineering team.
- **Agentic Workflow Systems —** Engineered LangChain/FlowiseAI agents for multi-step reasoning and tool use — designing tool interfaces, memory schemas, and failure-recovery logic. Instrumented agent traces for offline analysis of reasoning chains, which directly informed prompt and architecture revisions.
- **Dataset Quality & Model Performance Initiative —** Drove dataset auditing, labeling-rubric standardization, and inter-annotator agreement analysis. Identified systematic labeling errors and class imbalances through EDA, which translated directly into the ~20% accuracy improvement on downstream models.

EDUCATION

B. Tech in Computer Science - AI/ML, 2021 – 2024 | Ahmedabad, India

L.J Institute of Engineering and Technology

(8.3 CGPA)

SKILLS

Certifications
Building Deep learning models with TensorFlow (IBM), Introduction to computer vision and image processing (IBM), Introduction to Deep learning and Neural networks with keras (IBM), Data Analysis with python (IBM), Data Visualization (IBM), Python for Data Science, Al and Development (IBM), Database and SQL for Data science with python (IBM), GANs and Diffusion model in machine learning (LinkedIn Learning), Deep learning-Image processing (LinkedIn Learning), Microsoft Certified: Azure Fundamentals (Microsoft), Machine learning with python (IBM), python and data visualization (Kaggle)

Generative Al
Transformer-based architecture (GPT, BERT, T5), fine-tuning large language models (LLMs), Retrieval-Augmented Generation (RAG), Hugging Face, and prompt engineering and optimisation, FlowiseAI, RASA, LangChain, LangGraph, OpenAl Assistant API, Realtime API, Gemini API

Database
ChromaDB, PostgreSQL, MongoDB, MySQL

Machine Learning
Regression, classification, clustering, neural networks (ANN, CNN, RNN), natural language processing (NLP), model optimization, hyperparameter tuning.

Programming & Frameworks
Python, Keras, TensorFlow, PyTorch, Flask, FastAPI, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Javascript

DevOps & CI/CD
Docker, Docker Compose, GitHub Actions, AWS (EC2, S3, Lambda, ECS)

HONOURS & ACHIEVEMENTS

Professional Certification of Data Scientist, *IBM*